

The Role of Artificial Intelligence in Drug Discovery and Repurposing

A Mini Research Project

Prepared for: CodeAlpha Pharmaceutical Research Internship — Task 4

Domain: Pharmaceutical Research

Date: August 2026

1. Introduction

Traditional drug discovery is slow, expensive, and prone to failure: bringing a single new molecule from target identification to market approval typically takes ten to fifteen years and costs over a billion dollars, with the majority of candidates failing during preclinical or clinical testing. Over the last decade, artificial intelligence (AI) and its subfields — machine learning (ML), deep learning (DL), and natural language processing (NLP) — have moved from experimental curiosities to tools with genuine clinical impact, with AI-designed therapeutics now progressing through human trials across multiple disease areas.

This report examines two closely related applications of AI in pharmaceutical research: (i) AI-driven drug discovery, the use of computational models to identify and design entirely new therapeutic candidates, and (ii) AI-driven drug repurposing, the use of similar computational techniques to find new therapeutic uses for drugs that are already approved or in advanced clinical development. Both approaches aim to compress the drug development timeline, reduce cost, and improve the probability of clinical success by replacing slow, trial-and-error laboratory work with predictive, data-driven models.

2. The Traditional Drug Discovery Bottleneck

The conventional pipeline moves through several distinct stages: target identification and validation, hit discovery, lead optimization, preclinical safety testing, and three phases of clinical trials before regulatory review. Each stage acts as a filter, and the attrition rate is severe — the vast majority of compounds that enter early discovery never reach the market. Key bottlenecks include the sheer size of chemical space (estimated at more than 10^{60} possible drug-like molecules, far beyond what can be physically synthesized and tested), the difficulty of predicting how a molecule will behave in a living system from its structure alone, and the high cost of running clinical trials to test efficacy and safety in humans.

AI addresses these bottlenecks by replacing exhaustive physical experimentation with predictive computation: models can screen millions of virtual compounds *in silico*, prioritizing only the most promising candidates for laboratory synthesis and testing.

3. AI in Drug Discovery

3.1 Target Identification and Validation

Before a drug can be designed, researchers must identify a biological target — usually a protein — whose modulation will produce a therapeutic effect. AI models trained on genomic, proteomic, and clinical data can mine large biomedical datasets to flag disease-associated genes and proteins far faster than manual literature review. Natural language processing tools also scan millions of research papers and databases to surface previously overlooked target-disease associations.

3.2 Protein Structure Prediction

A major breakthrough in this area is AlphaFold, DeepMind's deep learning system for predicting three-dimensional protein structure from amino acid sequence. AlphaFold and related models have generated hundreds of millions of predicted protein structures, giving researchers structural insight into proteins that would otherwise take months or

years to solve experimentally through X-ray crystallography or cryo-electron microscopy. Knowing a target's 3D shape is essential for structure-based drug design, since it allows researchers to computationally model how a candidate molecule will physically bind to the target.

3.3 Virtual Screening and De Novo Molecule Design

Once a target is known, AI models can perform virtual screening — rapidly scoring large virtual libraries of compounds for their likely binding affinity to the target — to shortlist candidates for laboratory synthesis. A newer and more ambitious approach is generative chemistry, in which deep generative models (including generative adversarial networks and transformer-based architectures) design entirely novel molecular structures optimized for desired properties such as potency, selectivity, and low toxicity, rather than merely filtering existing compound libraries. Insilico Medicine's use of a generative adversarial network platform to identify a novel drug candidate within a matter of weeks, rather than years, is a widely cited demonstration of this approach; the company's PHD1 inhibitor for idiopathic pulmonary fibrosis, discovered using this pipeline, has since advanced into clinical trials.

3.4 Preclinical Safety and ADMET Prediction

AI models are also used to predict a candidate molecule's absorption, distribution, metabolism, excretion, and toxicity (ADMET) profile before it ever reaches an animal or human study. By flagging compounds likely to fail on safety or pharmacokinetic grounds early, these models help concentrate laboratory resources on candidates with a genuinely higher probability of clinical success.

4. AI in Drug Repurposing

Drug repurposing (also called drug repositioning) seeks new therapeutic uses for molecules that already have regulatory approval or have reached late-stage clinical testing. Because an approved drug's safety, toxicology, and manufacturing profile is already documented, repurposing can bypass much of the lengthy preclinical safety package required for a brand-new chemical entity, allowing sponsors to move more directly into efficacy-focused trials.

4.1 Computational Approaches

Two computational approaches dominate AI-assisted repurposing. Network medicine approaches map diseases and drugs onto shared biological pathways to identify unexpected mechanistic overlaps. Knowledge-graph approaches integrate structured data from scientific literature, clinical trial records, and drug databases into a single searchable graph, allowing researchers to query for drugs that act on mechanisms relevant to a new disease of interest.

4.2 Case Study: Baricitinib and COVID-19

The most frequently cited example of AI-assisted drug repurposing reaching full clinical validation is baricitinib, a Janus kinase (JAK) 1/2 inhibitor originally approved for rheumatoid arthritis. In January 2020, researchers at BenevolentAI used an AI-enhanced biomedical knowledge graph, combined with expert-guided queries, to map the likely mechanism of SARS-CoV-2 cell entry and identify approved drugs capable of interrupting it. Within roughly 48 hours of analysis, the workflow flagged baricitinib as a candidate with dual antiviral activity — through inhibition of AAK1 and related kinases involved in viral endocytosis — and anti-inflammatory activity, through JAK1/2-mediated suppression of cytokine signalling.

This computational prediction was subsequently validated in randomized controlled trials: the ACTT-2 Phase 3 trial and the COV-BARRIER trial both reported clinically meaningful benefits, including reduced mortality, when baricitinib was added to standard COVID-19 care. The drug went on to receive FDA emergency use authorization and a subsequent WHO recommendation for hospitalized COVID-19 patients. The case is widely used to illustrate how AI-guided hypothesis generation, combined with rigorous clinical validation, can compress a repurposing timeline that would traditionally take years into a matter of months.

4.3 Other Notable Examples

- Healx used AI-driven analysis to identify HLX-0201 as a candidate for fragile X syndrome, advancing the project to Phase II trials within roughly eighteen months.
- Deep Genomics applied an AI platform to identify novel oligonucleotide candidates for Wilson's disease, completing target-to-candidate work in about eighteen months.
- Broader knowledge-graph efforts compiled during the COVID-19 pandemic flagged dozens of approved drugs — including corticosteroids that were later confirmed effective in the RECOVERY trial — as potential repurposing candidates for further clinical testing.

5. Benefits of AI in Pharmaceutical Research

- Speed: Candidate identification that once took years can, in favorable cases, be compressed to weeks or months.
- Cost reduction: Fewer compounds need to be physically synthesized and tested, and repurposing bypasses much of the preclinical safety package.
- Improved success rates: Predictive ADMET and target-binding models help filter out likely failures before they reach costly clinical trials.
- Access to previously "undruggable" targets: Structure prediction tools such as AlphaFold provide structural insight into proteins that lack experimentally solved structures.
- Rapid response to emerging health threats: Repurposing pipelines can propose treatments for novel diseases, such as COVID-19, far faster than de novo drug development.

6. Limitations and Challenges

- Data quality and bias: AI models are only as reliable as the biomedical data they are trained on, which is often incomplete, inconsistently annotated, or skewed toward well-studied targets and diseases.
- False positives: Sparse or unvalidated interaction data can generate mechanistically plausible but clinically ineffective candidates, and every AI-generated hypothesis still requires laboratory and clinical validation.
- Interpretability: Deep learning predictions are often difficult to explain mechanistically, which complicates regulatory review and scientific trust.

- Regulatory and translational gap: An AI-designed molecule or repurposing hypothesis must still pass through the same clinical trial and approval process as any other candidate, so AI shortens early-stage timelines more than it shortens clinical development.
- Intellectual property and reproducibility questions: Proprietary AI platforms and datasets can make it difficult for independent researchers to verify or reproduce reported results.

7. Future Outlook

The AI-in-pharmaceuticals market has grown rapidly and is projected to continue expanding sharply over the coming years, reflecting both increased investor confidence and a growing number of AI-originated candidates entering clinical trials. Future progress is likely to come from combining multiple data modalities (genomic, structural, clinical, and real-world data) into unified foundation models, from closer integration between AI predictions and automated laboratory experimentation ("self-driving labs"), and from continued refinement of generative chemistry techniques capable of designing molecules with multiple optimized properties simultaneously. As regulatory agencies develop clearer frameworks for evaluating AI-derived evidence, AI is expected to move from an early-discovery accelerant to a tool embedded across the full drug development lifecycle.

8. Conclusion

Artificial intelligence has demonstrably reshaped early-stage pharmaceutical research, accelerating target identification, enabling de novo molecule design, and — as the baricitinib case illustrates — allowing existing drugs to be redeployed against new diseases within a fraction of the traditional timeline. While AI has not eliminated the fundamental cost and risk of clinical development, it has meaningfully compressed the discovery phase and improved the odds that a candidate entering clinical trials will succeed. Realizing AI's full potential will depend on better data quality, greater model interpretability, and continued rigorous clinical validation of AI-generated hypotheses.

References

1. Ferreira, F. J. N., & Carneiro, A. S. (2025). AI-Driven Drug Discovery: A Comprehensive Review. ACS Omega. <https://doi.org/10.1021/acsomega.5c00549>
2. Dharmasivam, M., Azad, M. G., Richardson, V., Kaya, B., & Richardson, D. R. (2026). Artificial intelligence–assisted drug discovery in 2025: Faster, but is it better? *Pharmacological Reviews*, 78(1), 100103.
3. Vemula, D., et al. (2024). Artificial Intelligence for Drug Discovery: Are We There Yet? *Annual Review of Pharmacology and Toxicology*, 64, 527–550. <https://doi.org/10.1146/annurev-pharmtox-040323-040828>
4. Author(s) not individually specified (2025). The future of pharmaceuticals: Artificial intelligence in drug discovery and development. *Journal of Pharmaceutical Analysis*, 15(8), 101248. <https://doi.org/10.1016/j.jpha.2025.101248>
5. Smith, D. P., Oechsle, O., Rawling, M. J., Savory, E., Lacoste, A. M. B., & Richardson, P. J. (2021). Expert-Augmented Computational Drug Repurposing Identified Baricitinib as a Treatment for COVID-19. *Frontiers in Pharmacology*, 12, 709856. <https://doi.org/10.3389/fphar.2021.709856>

6. Richardson, P., et al. (2022). The AI-Assisted Identification and Clinical Efficacy of Baricitinib in the Treatment of COVID-19. (PMC review article).
7. AI-enabled drug and molecular discovery: computational methods, platforms, and translational horizons. (2025). Discover Molecules, Springer Nature. <https://doi.org/10.1007/s44345-025-00037-5>
8. Review: Artificial intelligence revolution in drug discovery. (2025). ScienceDirect.
9. Drug Discovery News (2025). AI-powered drug repurposing: Finding new uses for existing molecules. <https://www.drugdiscoverynews.com>